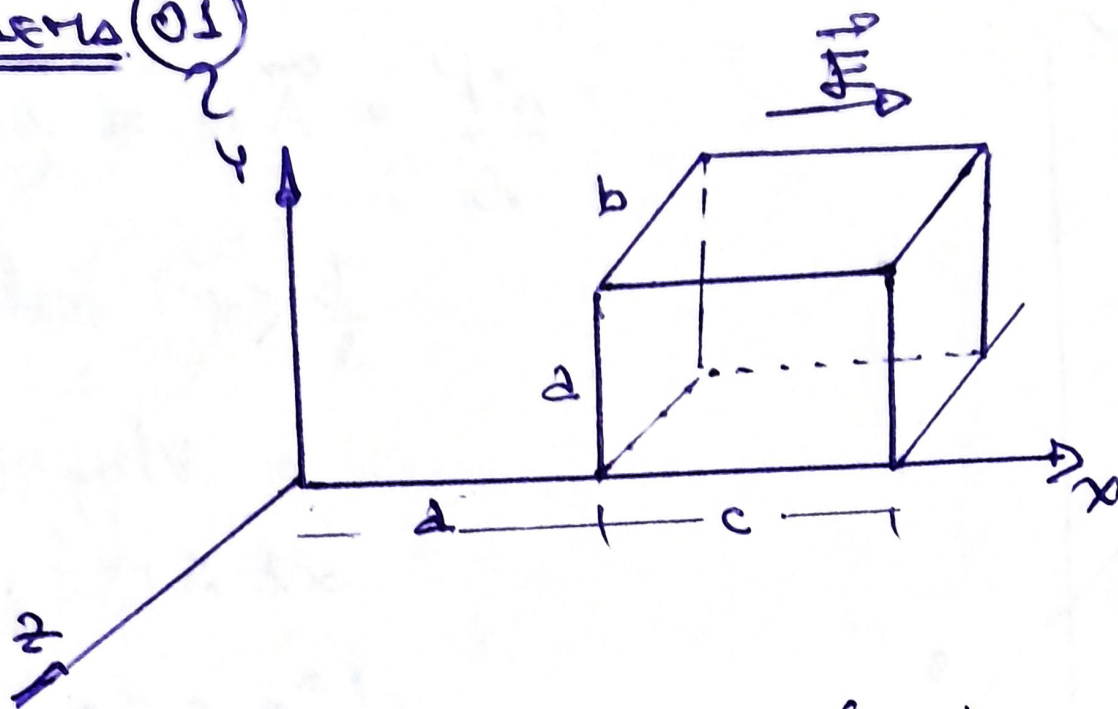


Física II
PC 02

Problema 01



$$\Phi_E = -\left(E_x \Big|_{x=a}\right) A + \left(E_x \Big|_{x=a+c}\right) A$$

$$\Phi_E = -(3+2a^2)ab + (3+2(a+c)^2)ab$$

$$\Phi_E = 2abc(2a+c)$$

REEMPLAZANDO LOS VALORES

$$\Phi_E = 0,269 \text{ N} \cdot \text{m}^2/\text{C}$$

Problema 02

Usaremos la ley de Gauss

$$\oint \vec{E} \cdot d\vec{A} = \frac{q_{in}}{\epsilon_0}$$

a) Para $x > \frac{d}{2}$

$$dq = \rho dV$$

$$dq = \rho A dx$$

$$dq = CA x^2 dx$$

$$\therefore \int \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} dq$$

$$EA = \frac{CA}{\epsilon_0} \int_0^{d/2} x^2 dx$$

$$EA = \frac{1}{3} \left(\frac{CA}{\epsilon_0} \right) \left(\frac{d^3}{8} \right)$$

$$E = \frac{Cd^3}{24\epsilon_0}$$

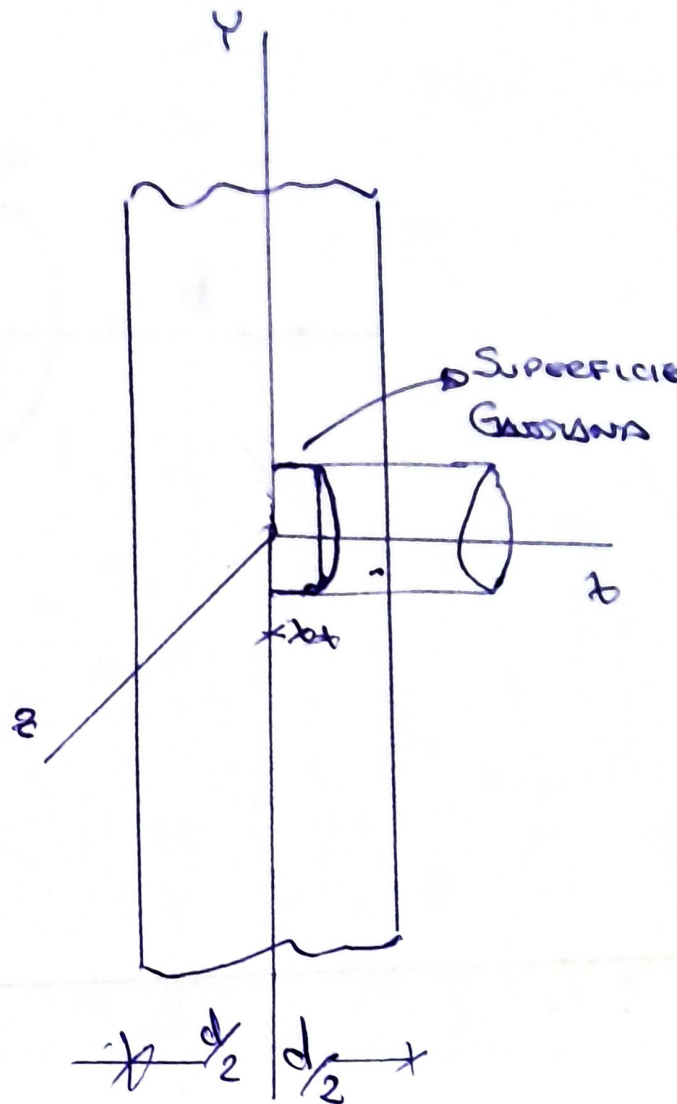
o) Para $x > \frac{d}{2}$

$$E = \frac{Cd^3}{24\epsilon_0} ;$$

Para $x < \frac{d}{2}$

Para $x < -\frac{d}{2}$

$$E = -\frac{Cd^3}{24\epsilon_0}$$



b) Para $-\frac{d}{2} < x < \frac{d}{2}$

$$\int \vec{E} \cdot d\vec{A} = \frac{1}{\epsilon_0} \int dq$$

$$EA = \frac{CA}{\epsilon_0} \int_0^x x^2 dx$$

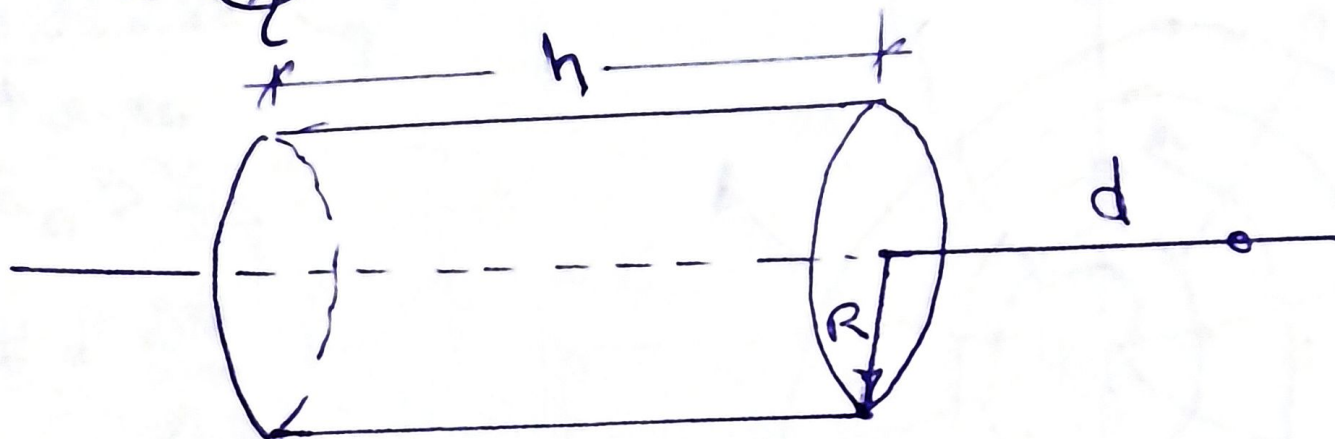
Para $x > 0$

$$E = \frac{Cx^3}{3\epsilon_0}$$

Para $x < 0$

$$E = -\frac{Cx^3}{3\epsilon_0}$$

Problems (03)



Area $dx \rightarrow$ Charge $\frac{Q dx}{h}$

$$dV = \frac{k Q dx}{h \sqrt{x^2 + R^2}}$$

$$V = \int dV$$

$$V = \int_d^{d+h} \frac{k Q dx}{h \sqrt{x^2 + R^2}}$$

$$V = \frac{k Q}{h} \ln(x + \sqrt{x^2 + R^2}) \Big|_d^{d+h}$$

$$V = \frac{k Q}{h} \ln \left(\frac{d+h + \sqrt{(d+h)^2 + R^2}}{d + \sqrt{d^2 + R^2}} \right)$$

Problema (04)

Figura

a) $E_A > E_B$

$$E = \frac{\Delta V}{\Delta S}$$

b) $E_B = - \frac{\Delta V}{\Delta S}$

$$E_B = - \frac{(6-2) \text{ V}}{2 \text{ cm}}$$

$$E_B = 200 \text{ N/C}$$

↓
Hacia abajo

c) LAS LINEAS DE CAMPO
SE MUESTRAN EN LA FIGURA 1

